

F_{ST} - d_{XY} Correlation Pipeline: Methods

Analysis pipeline:

1. Shell script

- a. Filter VCF (VCFTools)
- b. Index VCF (BCFTTools)
- c. Convert VCF to Zarr (vcf2zarr)

2. Python script

- a. Convert Zarr to Xarray Dataset (sgkit)
- b. Group Dataset by cohort (population), split by chromosome, window by variant, chunk variants by window size (sgkit)
- c. Compute differentiation (F_{ST}) and divergence (d_{XY}) (sgkit)
- d. Compute Spearman's ρ (SciPy)
- e. Plot (matplotlib)

F_{ST} - d_{XY} Correlation Pipeline: Results

- Spearman's ρ was estimated with significance for all groups
- On chromosome Z, pairwise comparisons to the Lesina cohort yielded high ρ
 - High divergence $\sim$ high differentiation (cohort different from compared cohort relative to inbreeding)

